

Elvira Gouatieu Dongmo

Seminar TUD

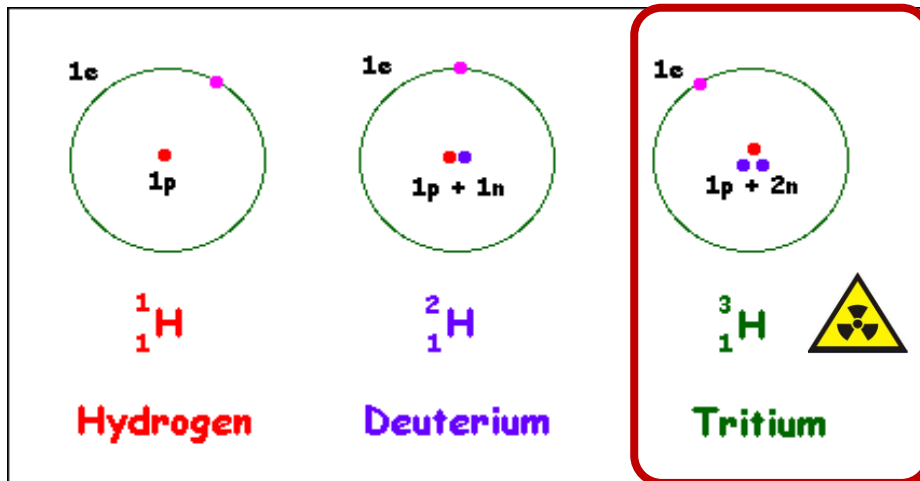
Strong anharmonicity in the vibrational spectra and numerical evaluation of the ZPE in a model nanopore

Theoretical Chemistry Seminar // October 04, 2022

Hydrogen isotopes and applications

Why hydrogen isotopes?

Hydrogen isotopes



<https://gifer.com/en/gifs/hydrogen>

Science



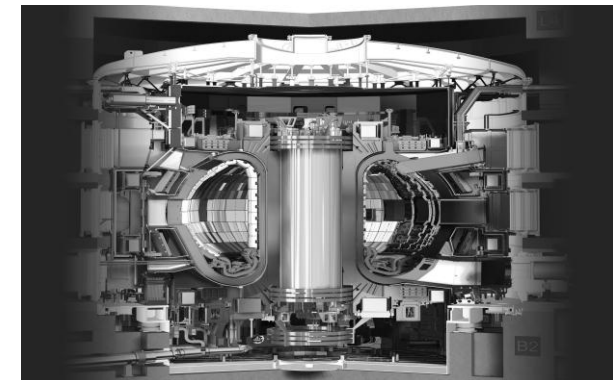
stuff.co.nz/stuff-nation/125940471

Medicine



theguardian.com/society

Nuclear fusion



iter.org/mach

Motivation



Numerical investigation

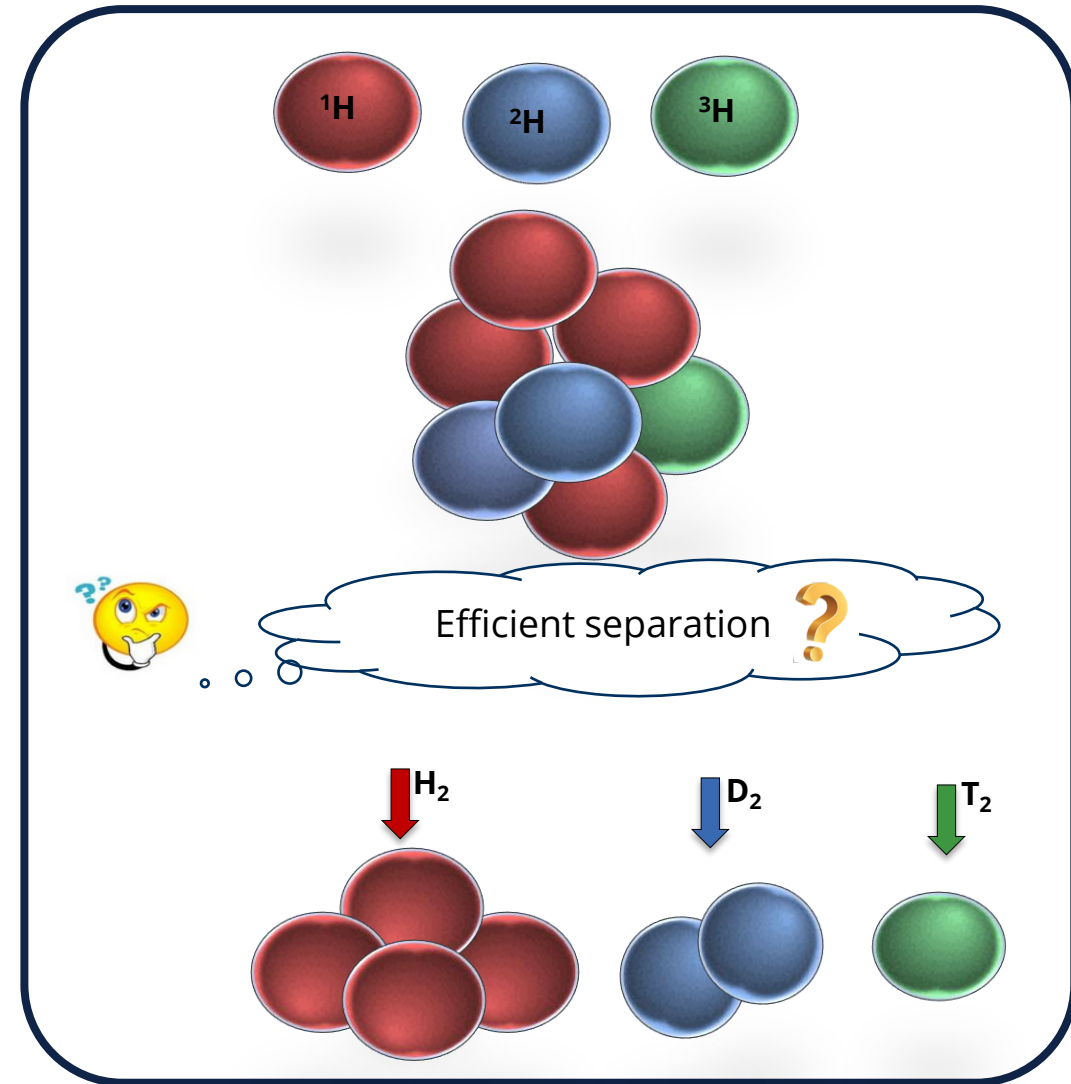


Experimental
research



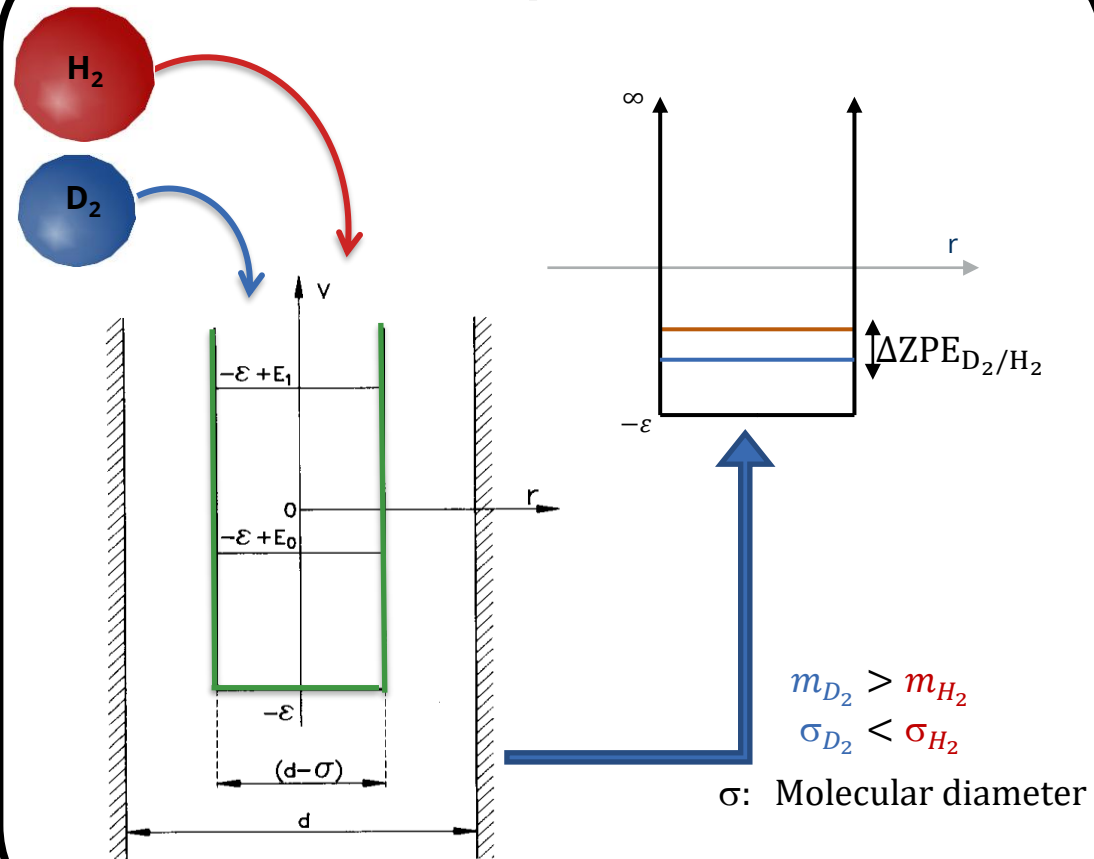
TECHNISCHE
UNIVERSITÄT
DRESDEN

HZDR



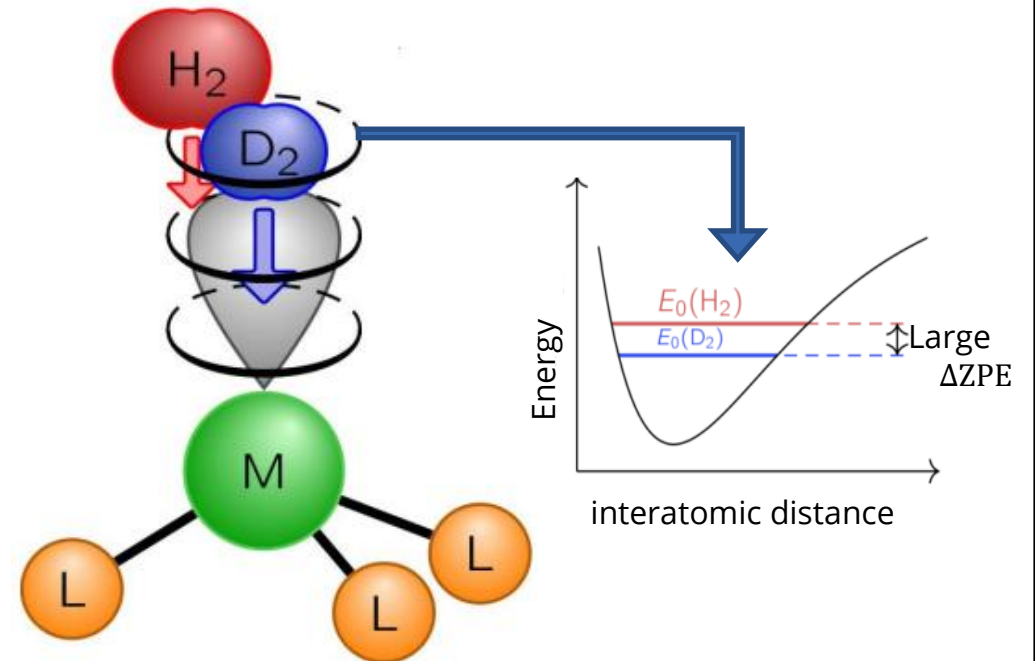
Approaches

Selective pore diffusion



J.J.M. Beenakker et al, Chem. Phys. Lett. 232 , 1995, 379–382.

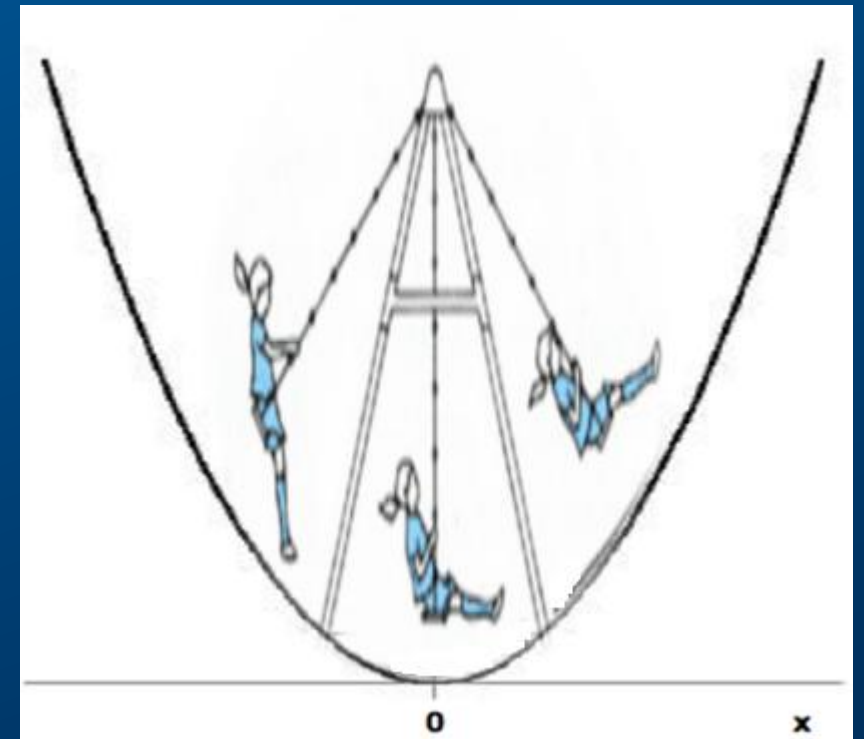
Strong adsorption site



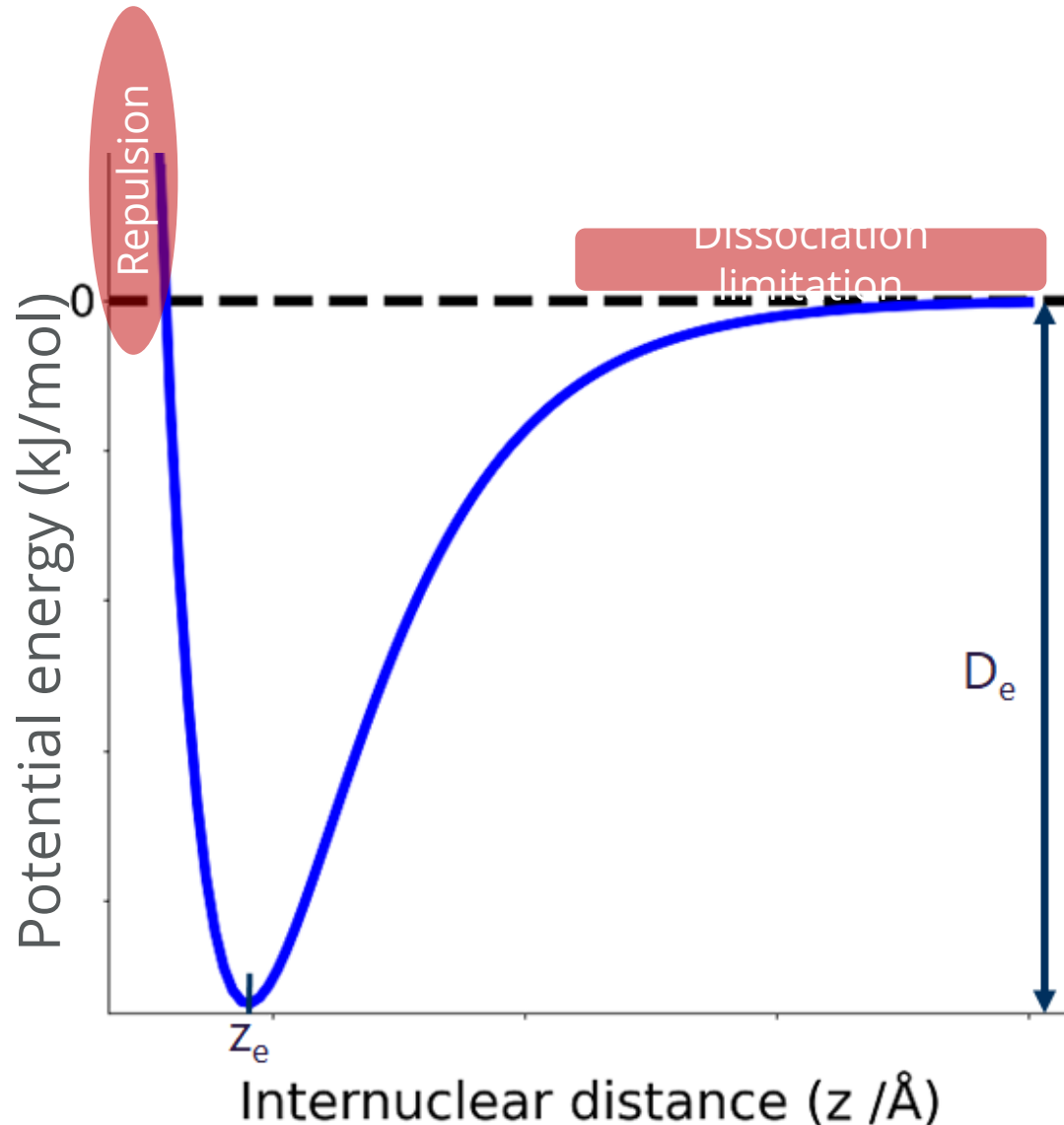
Toshiki Wulf, Thomas Heine, Int J Quantum Chem., 2018.



Beyond the harmonic oscillator model



Anharmonicity and Morse potential



□ Reality: anharmonic potential

✓ Good approximation: **Morse potential**

$$V(z) = D_e \left[\left(1 - e^{-\alpha(z-z_e)} \right)^2 - 1 \right]$$

✓ Small displacements from z_e

D_e : dissociation energy, z_e : bond length, α : constant parameter

Harmonic and anharmonic vibrational spectra

□ Taylor expansion

Set: $Z = (z - z_e)$

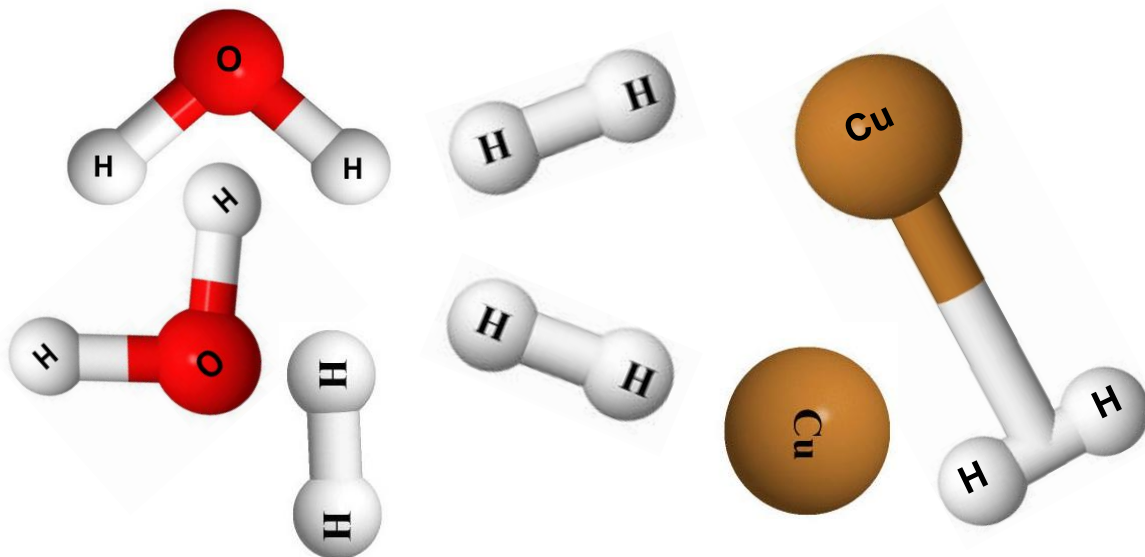
$$V(z) \approx V_0 + \underbrace{\left. \frac{dV(z)}{dz} \right|_{z=z_e}}_{=0} (Z) + \frac{1}{2} \left. \frac{d^2V(z)}{dz^2} \right|_{z=z_e} (Z)^2 + \frac{1}{6} \left. \frac{d^3V(z)}{dz^3} \right|_{z=z_e} (Z)^3 + \frac{1}{24} \left. \frac{d^4V(z)}{dz^4} \right|_{z=z_e} (Z)^4 + \dots$$

harmonic contributions anharmonic contributions

□ Energy correction

$$E_{\text{vib}}^{\text{anh}} = \underbrace{h\nu_o \left(n + \frac{1}{2} \right)}_{\text{harmonic}} - \underbrace{\frac{[h\nu_o (n + 1/2)]^2}{4D_e}}_{\text{anharmonic}}$$

n : quantum number
 ν_o : vibrational quantum number at the ground state
 h : Planck constant

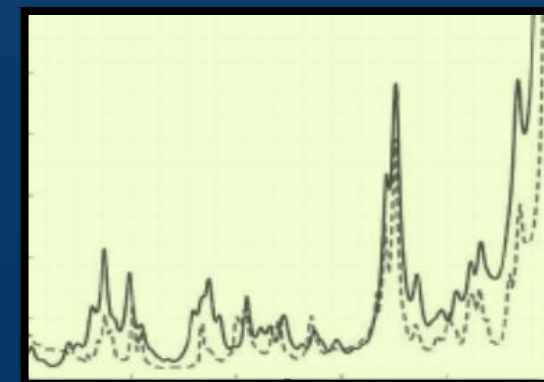


$$\mathbf{H}_{el}\Psi = E\Psi$$

VPT2

MP2

CCSD(T)

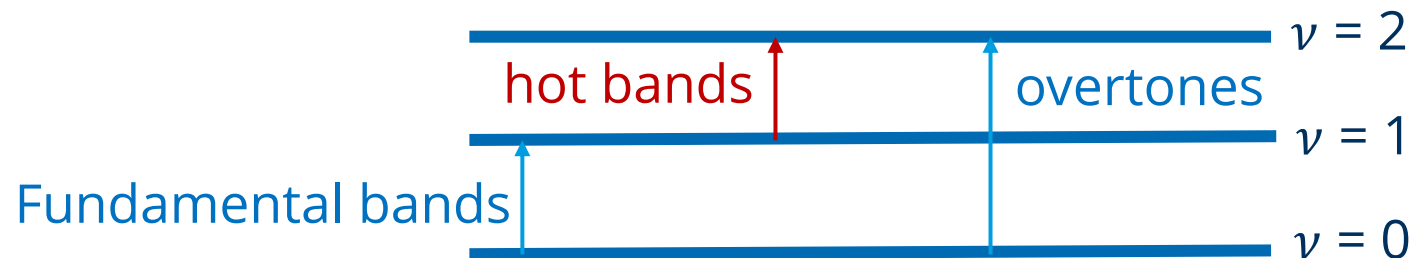


Strong anharmonicity in the vibrational spectra

Second-order vibrational perturbation theory (VPT2)

□ VPT2 has been extensively used in the past for:

- ✓ Vibrational energies and transition intensities^[1]
- ✓ To compute overtones and combination bands^[2]



□ Strength of the VPT2 approach:

- ✓ Cost efficiency to compute accurate vibrational properties
- ✓ Allows to compare directly the simulation and experimental results

Methods

- ✓ Expansion of the potential energy operator $\hat{V}(q)$

$$\hat{V}(q) \approx \frac{1}{2} \sum_i \omega_i q_i^2 + \frac{1}{6} \sum_{ijk} k_{ijk} q_i q_j q_k + \frac{1}{24} \sum_{ijkl} k_{ijkl} q_i q_j q_k q_l$$

Where:

$$k_{ijk} = \frac{\partial^3 V}{\partial q_i \partial q_j \partial q_k} \quad \text{and} \quad k_{ijkl} = \frac{\partial^4 V}{\partial q_i \partial q_j \partial q_k \partial q_l}$$

- ✓ Vibrational energy levels

$$E_v = \gamma_0 + \sum_i \omega_i \left(v_i + \frac{1}{2} \right) + \sum_i \sum_{j < i} \gamma_{ij} \left(v_i + \frac{1}{2} \right) \left(v_j + \frac{1}{2} \right)$$

anharmonic
contributions

γ_{ij} : set of anharmonic constants

ω_i : wavenumber

q : dimensionless normal coordinates

v : vibrational states

i, j, k and l are modes

Computational cost grows quickly with the system size

Example: $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_2$

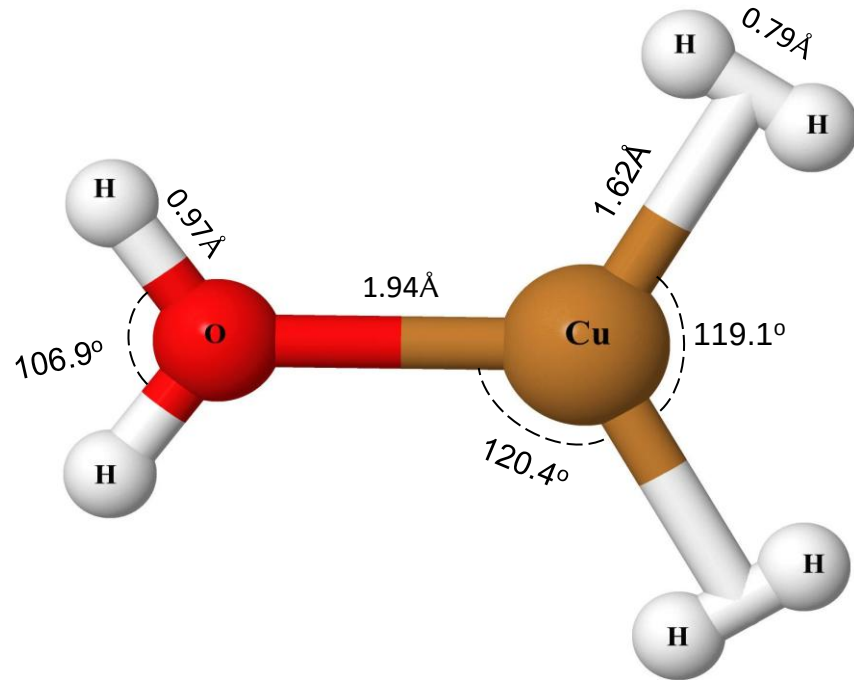


Figure 1: Optimized geometry of $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_2$ at the MP2/aug-cc-pVTZ level. The structure is planar with C_{2v} symmetry.

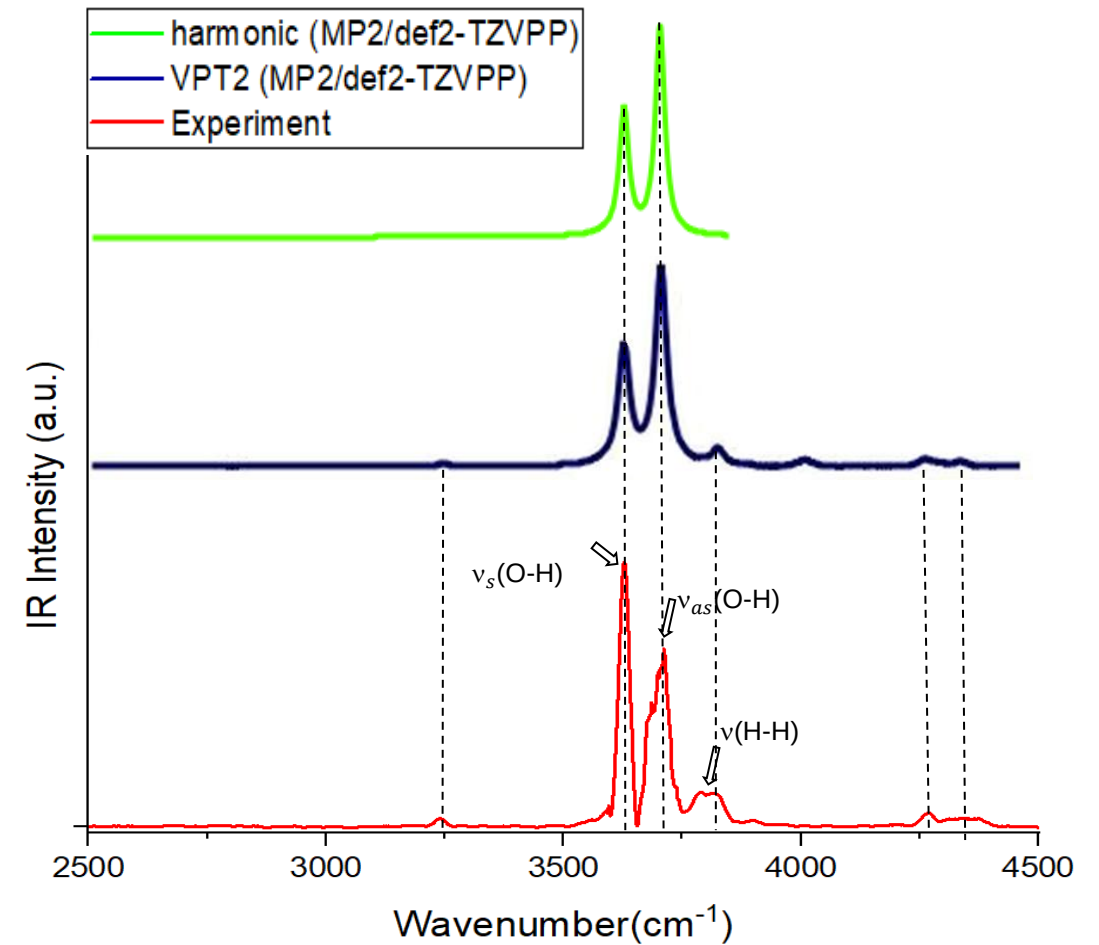
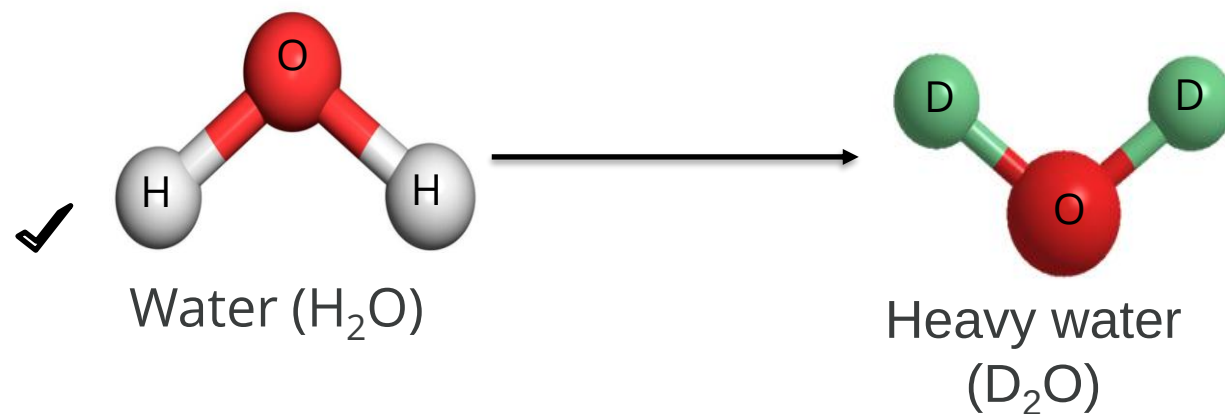
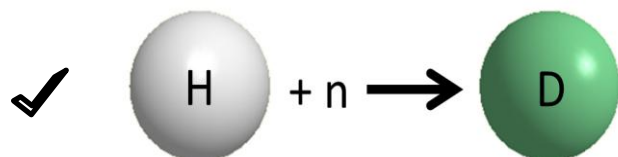


Figure 2: Comparison between experimental and computed vibrational spectra of $\text{Cu}^+(\text{H}_2\text{O})(\text{H}_2)_2$ in the O-H stretching frequency region.

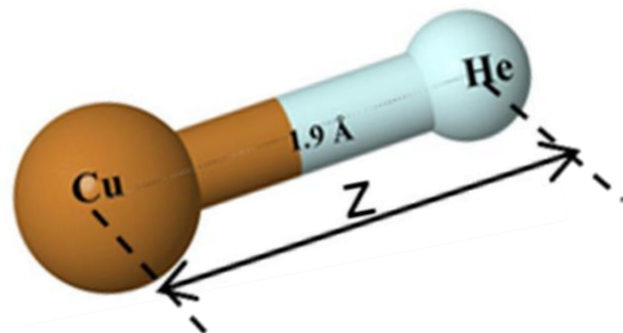
Ongoing...



Using VPT2 at the MP2/def2-TZVPP for:

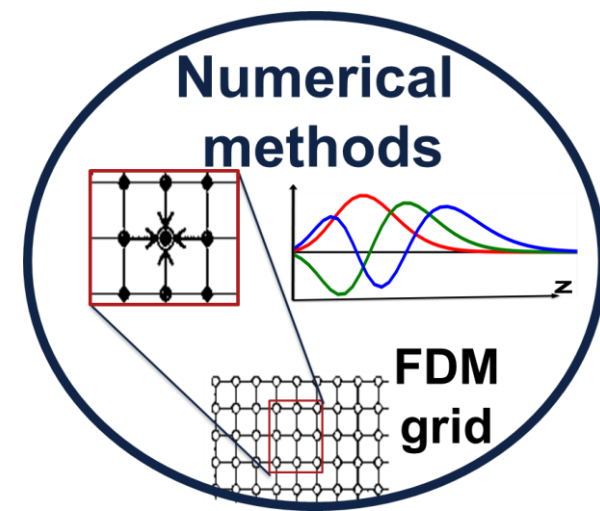
Showing up the vibrational behavior and H-H and D-D stretching bands in $\text{Cu}^+(\text{D}_2\text{O})(\text{H}_2)_2$, $\text{Cu}^+(\text{D}_2\text{O})(\text{D}_2)_2$, $\text{Cu}^+(\text{HDO})(\text{D}_2)_2$ complexes using CFOUR program

$$H_{\text{nuc}} \Psi = E \Psi$$



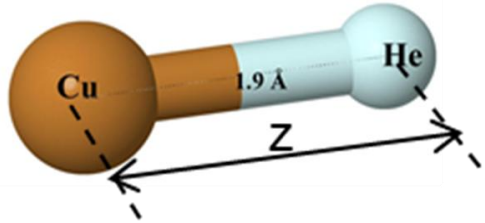
TIME INDEPENDENT
SCHRODINGER EQUATION

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V \right] \Psi = E \Psi$$



Numerical evaluation of the ZPE in a model nanopore
:
Investigation of He adsorbed at a metal center

Interaction potential energy



❖ Good approximation: **Fitted Morse potential**

$$V_{\text{Morse}}(z) = D_e \left[\left(1 - e^{-\alpha(z-z_e)} \right)^2 - 1 \right] \longrightarrow$$

D_e : dissociation energy, z_e : bond length,
 α : constant parameter

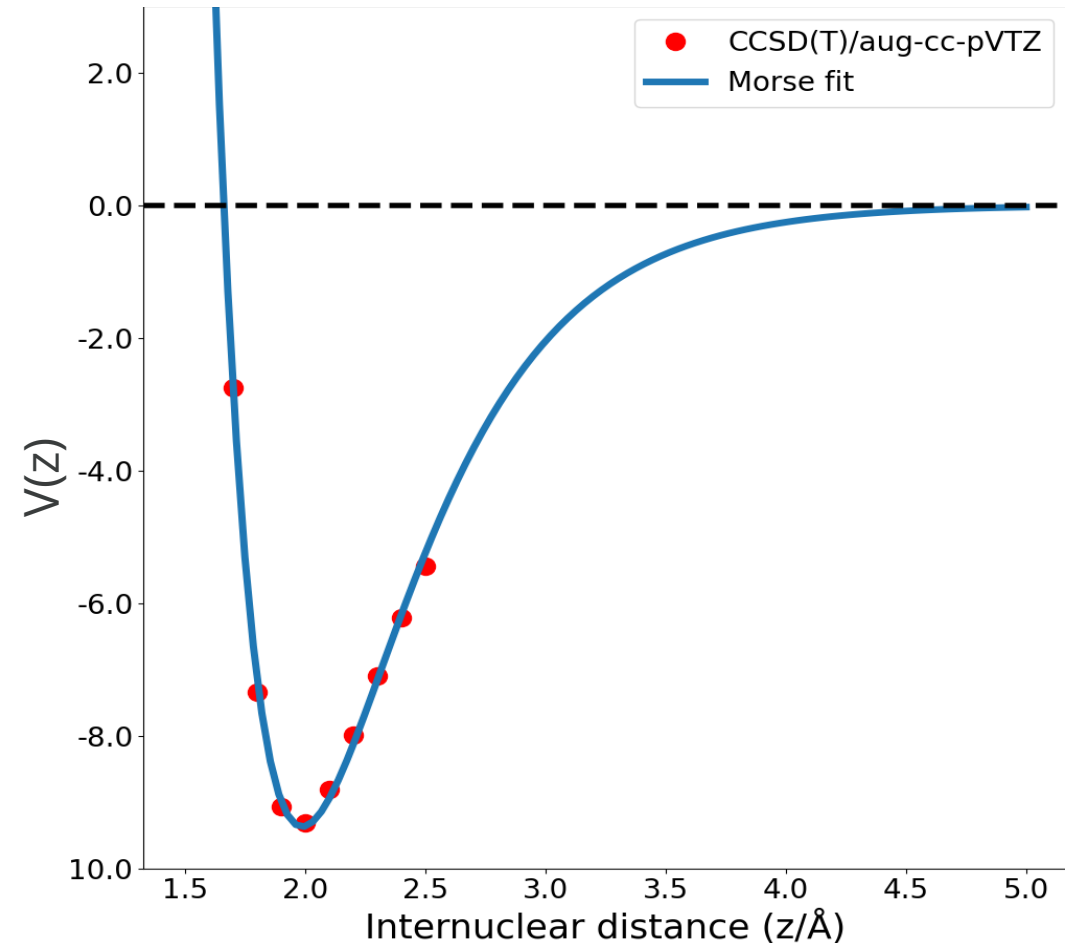


Figure 3: Fitted Morse potential at the CCSD(T)/aug-cc-pVTZ level of theory $D_e=9.37\text{kJ}\cdot\text{mol}^{-1}$, $\alpha = 2.12\text{\AA}^{-1}$, $z_e=1.9\text{\AA}$

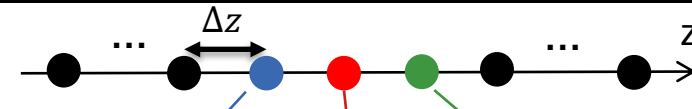
1D time independant Schrödinger equation for the Morse potential

$$\text{Schrödinger equation : } \hat{H}\Psi = E\Psi$$

- ✓ **ab initio calculations:** PES, CCSD(T), BSSE, CBS, ...
- ✓ **Analytical method:** ❖ Separation of variables
❖ zeros of Bessels functions
- ✓ **Numerical method :** Finite difference method(FDM)

$$\hat{H} = -\frac{\hbar^2}{2m^*} \frac{d^2}{dz^2} + V(z)$$

❖ 1D


$$\frac{d^2\Psi(z_i)}{dz^2} = \frac{\Psi(z_i - h) - 2\Psi(z_i) + \Psi(z_i + h)}{(\Delta z)^2}$$

Numerical solution

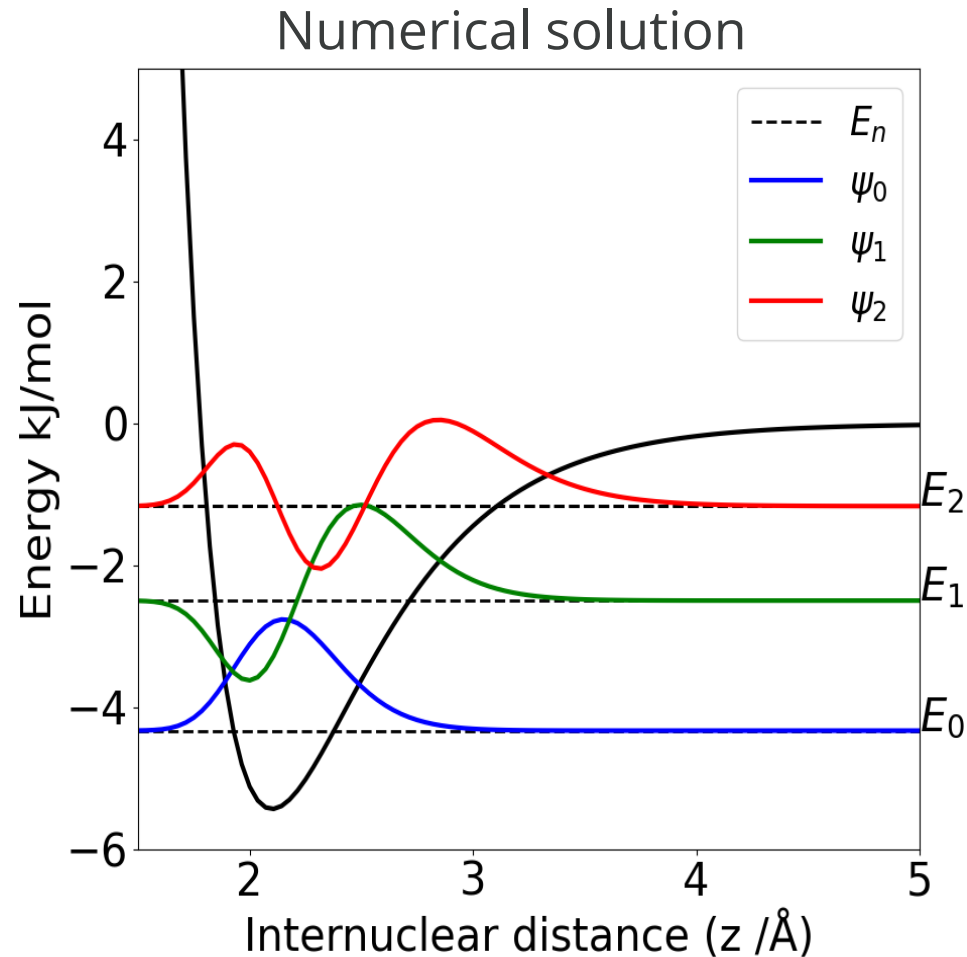


Figure 4: Numerical solution of the Schrödinger equation with the Morse potential at the **CCSD(T)/cc-pVTZ** level.

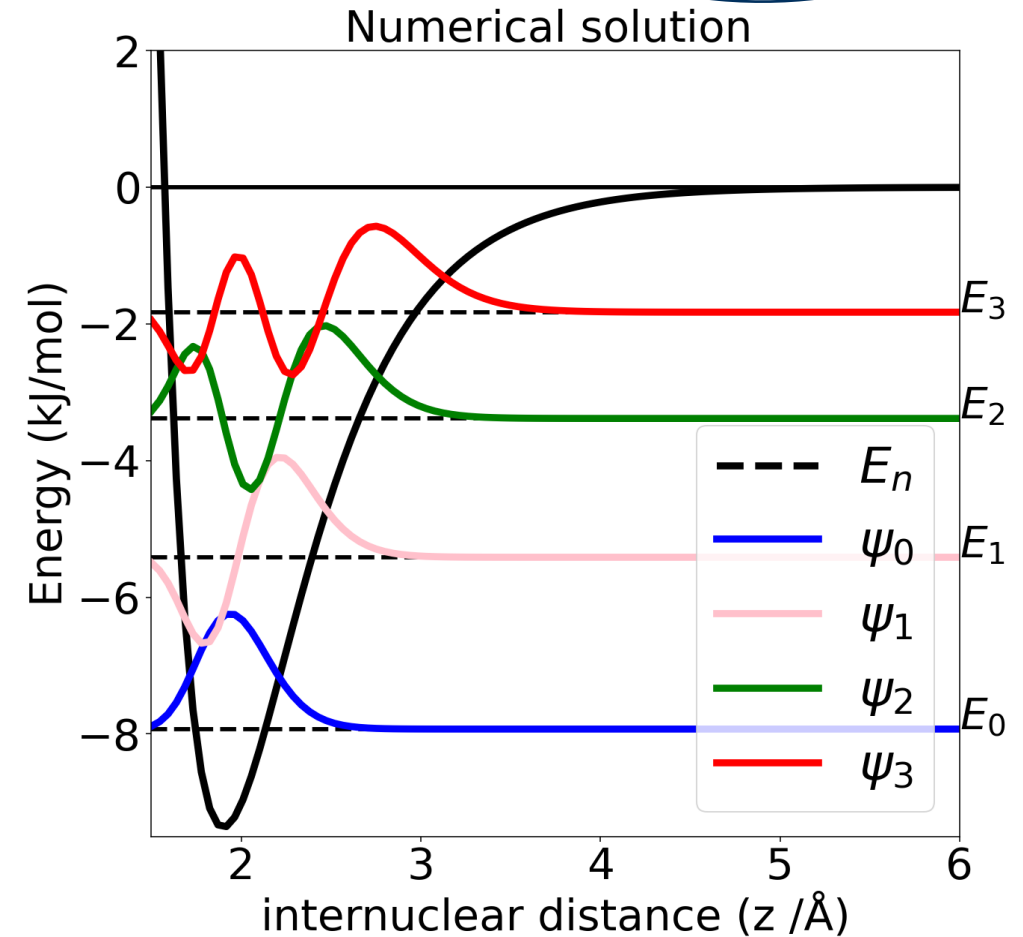


Figure 5: Numerical solution of the Schrödinger equation with the Morse potential at the **CCSD(T)/aug-cc-pVTZ** level.

Comparison

Table 1: Short analysis of the results : At the CCSD(T)/cc-pVTZ level of theory with the Morse parameters ($D_e=5.43 \text{ kJ. mol}^{-1}$, $\alpha = 2.16 \text{ \AA}^{-1}$, $z_e=2.1 \text{ \AA}$)

Energy levels	Numerical results (kJ. mol ⁻¹)	Analytical results (kJ. mol ⁻¹)	Relative errors
E_0	-4.3265	-4.3261	0.0093
E_1	-2.4956	-2.4941	0.0601
E_2	-1.1653	-1.1635	0.1545

Zero point energy difference

□ ZPE difference

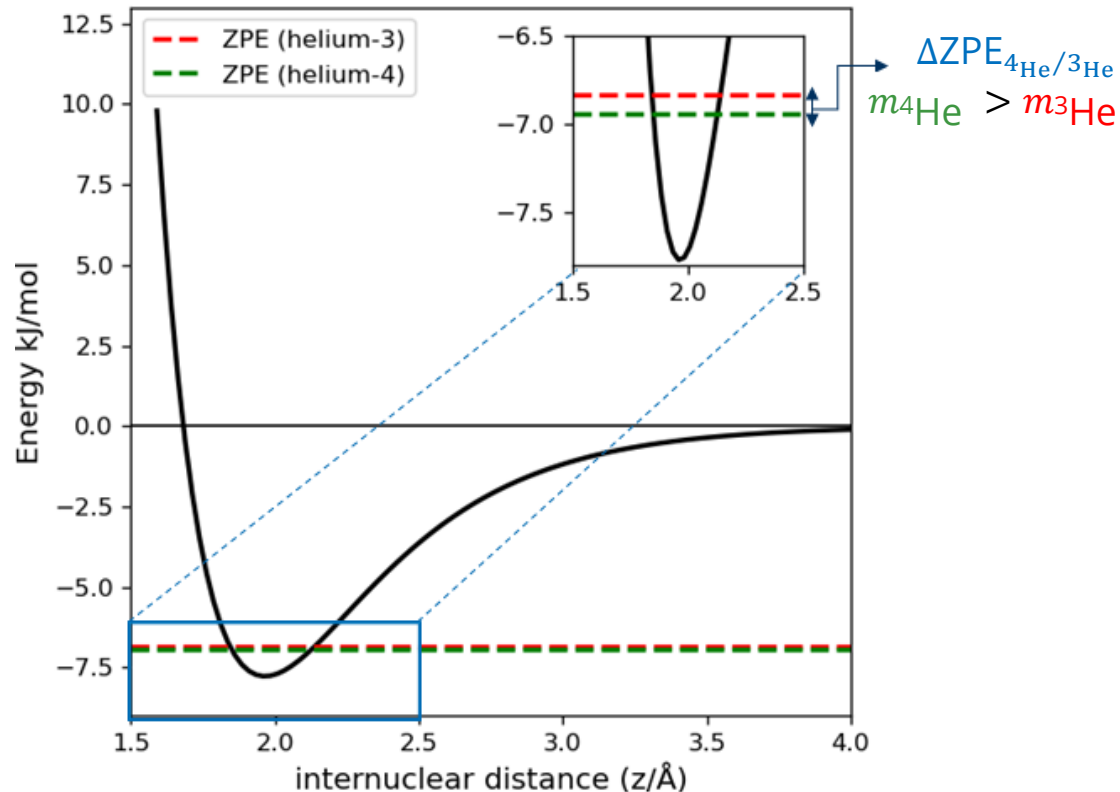
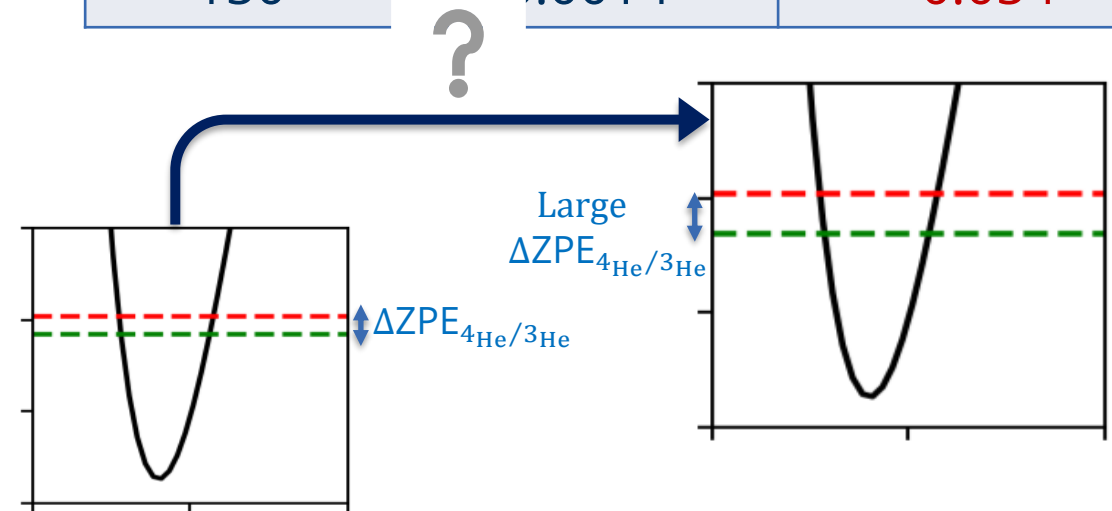


Figure 6: Numerical solution of the ZPE difference at the CCSD(T)/aug-cc-pVTZ level.

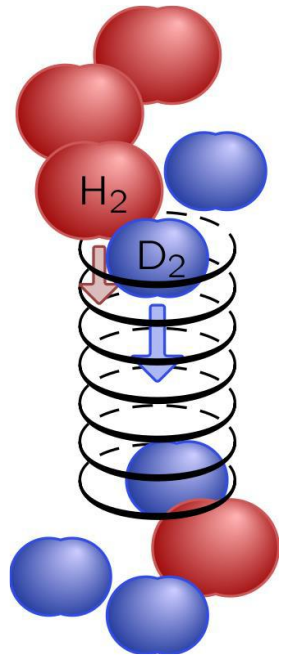
□ Accuracy and time complexity

Grid points (N)	Error of approximation	Computational time (secs)
100	0.0032	0.044
150	0.0014	0.054



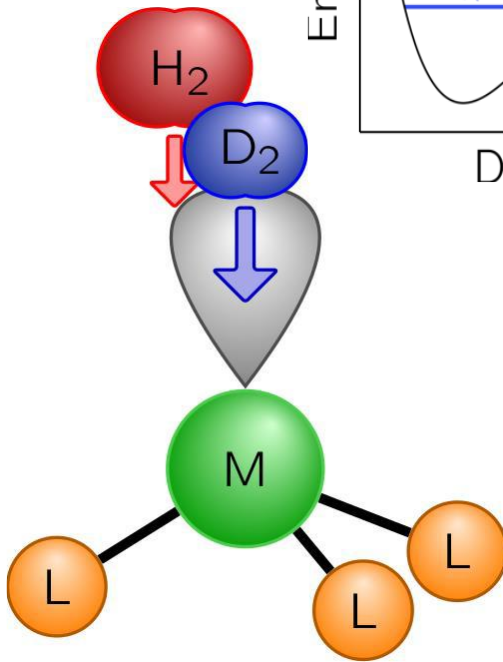
How to increase the ZPE difference?

State of the art

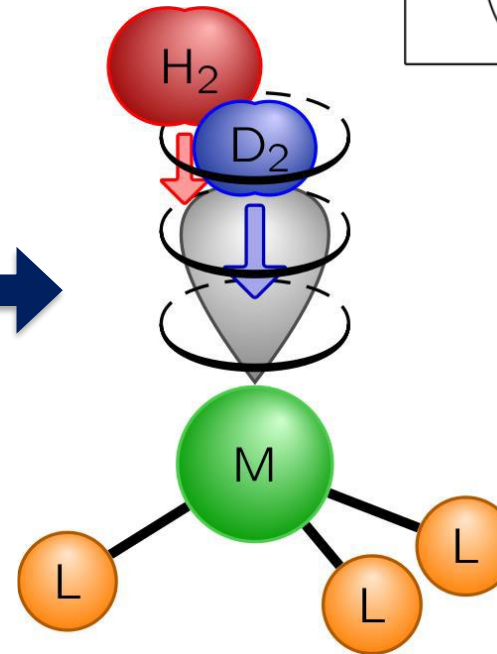
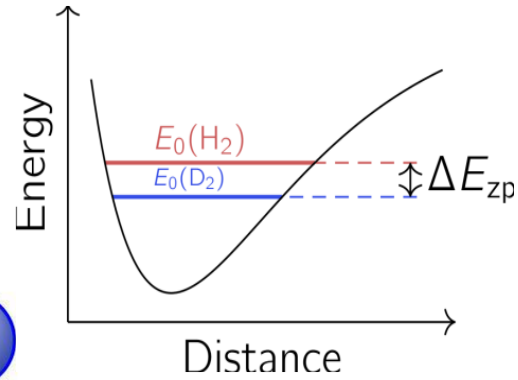


Selective pore
diffusion

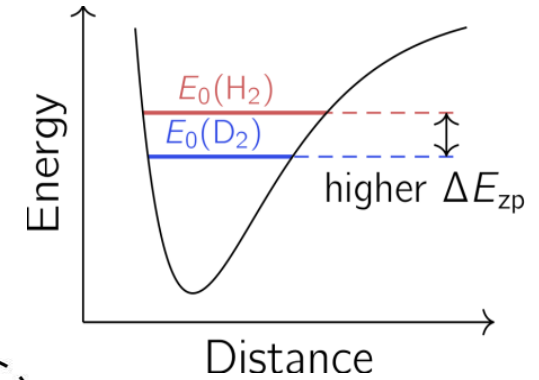
+



Chemical
affinity
quantum
sieving

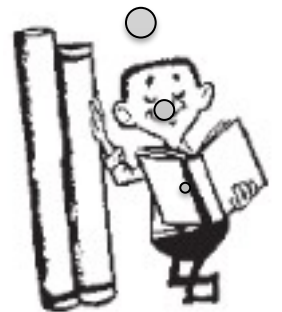


Selective
adsorption + pore
confinement

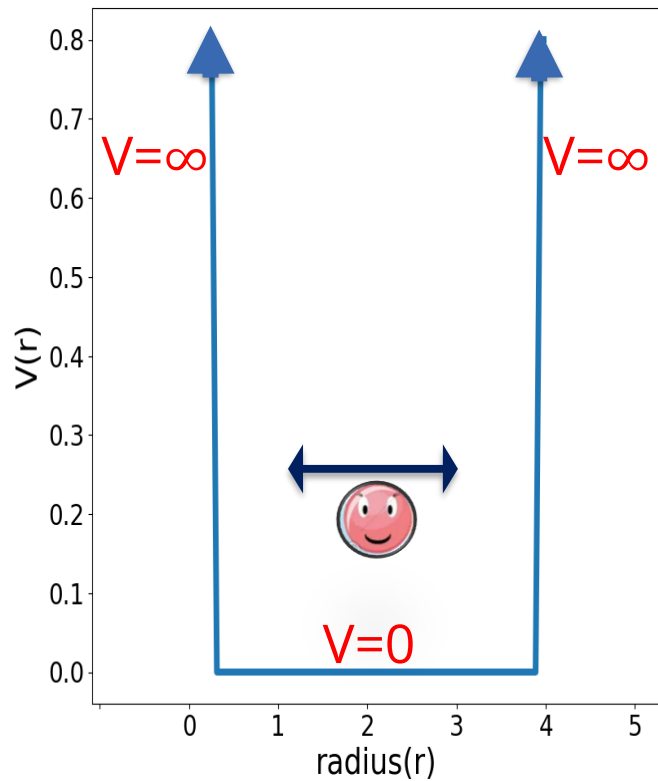


How to
implement
it?

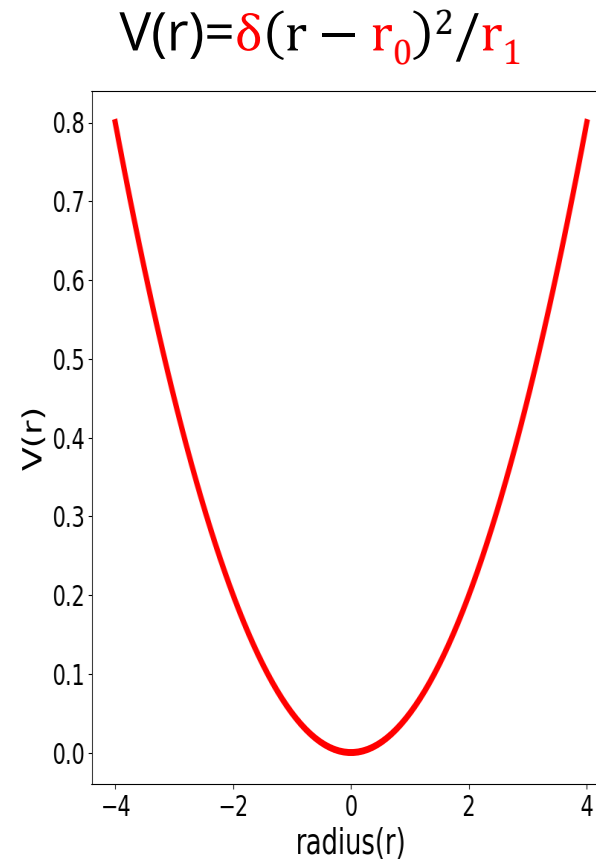
How to apply
it for ^3He and
 ^4He ?



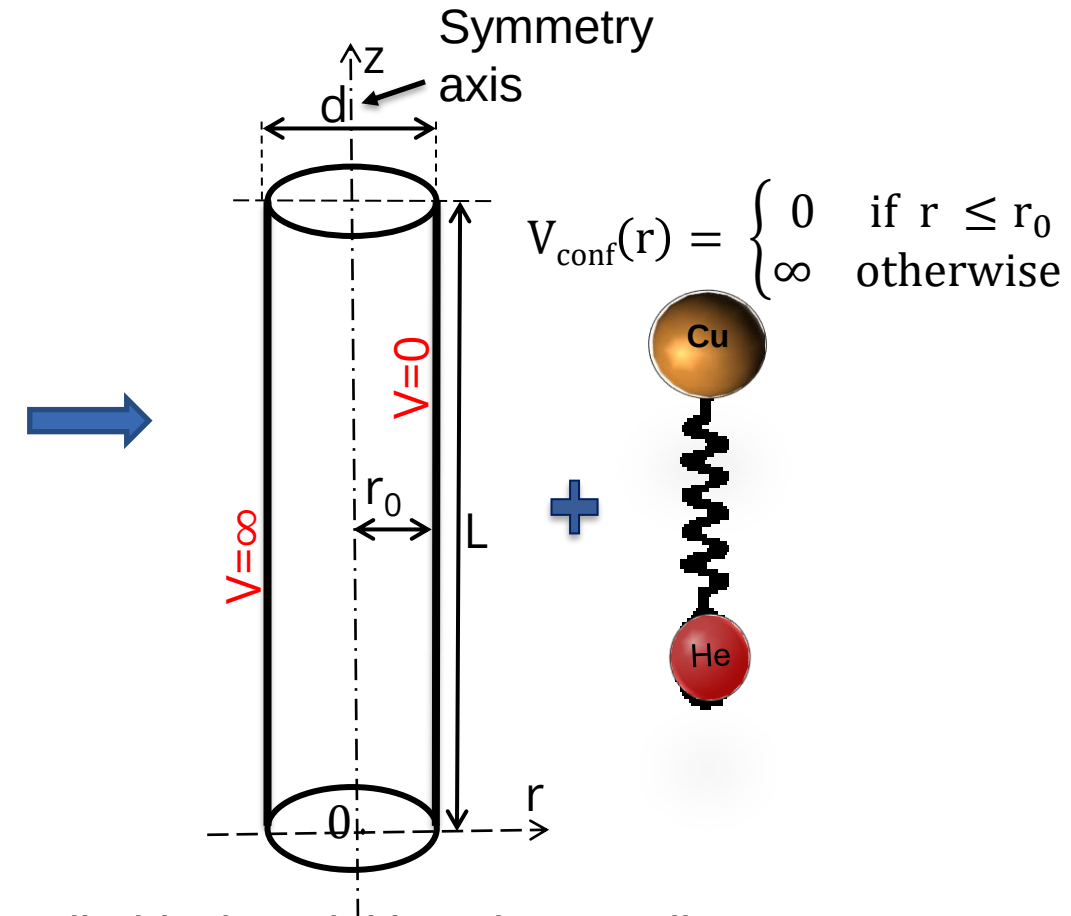
Potential profile



Hard wall model

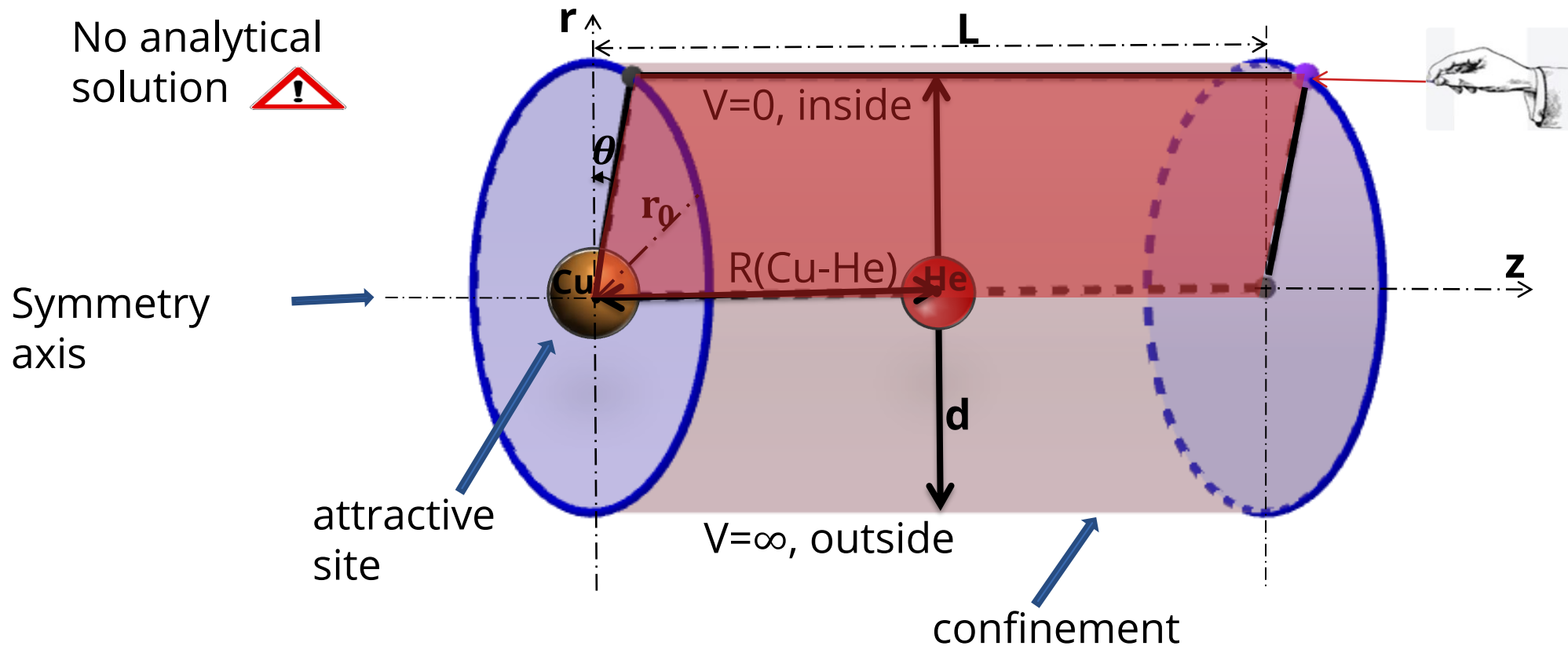


Harmonic oscillator model

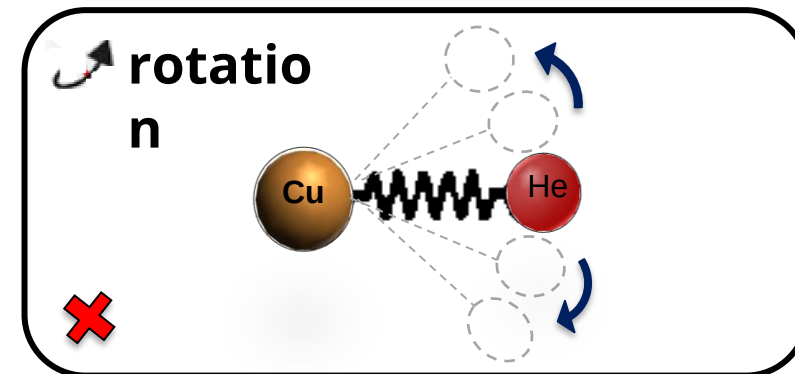
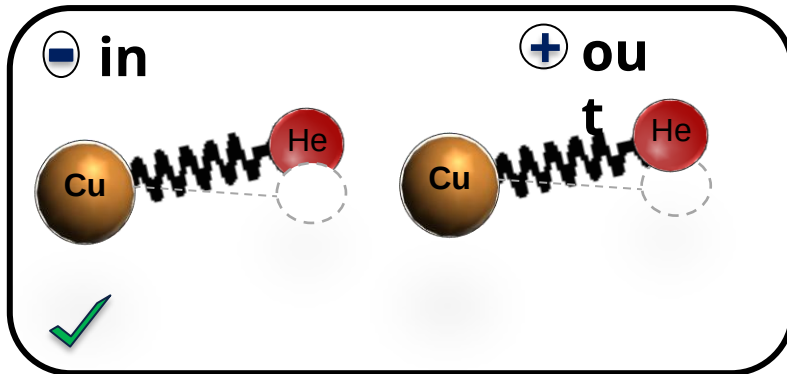
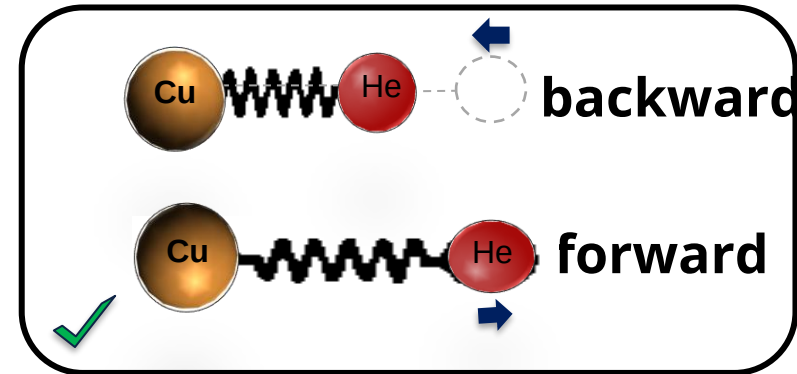
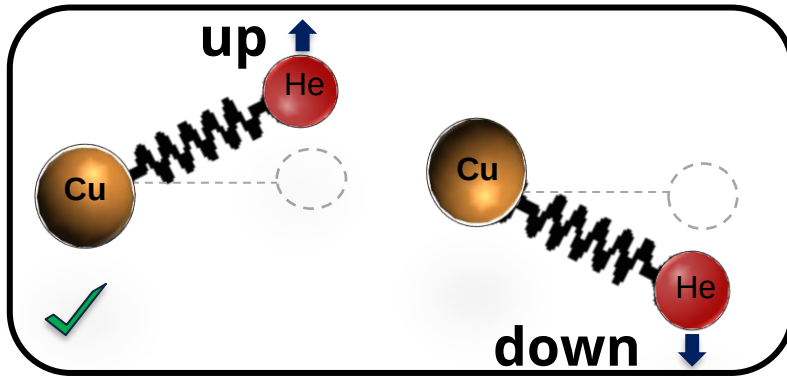


cylindrical model in polar coordinates (r, θ, z) with no variation in the θ direction

Strong adsorption site + cylindrical pore confinement

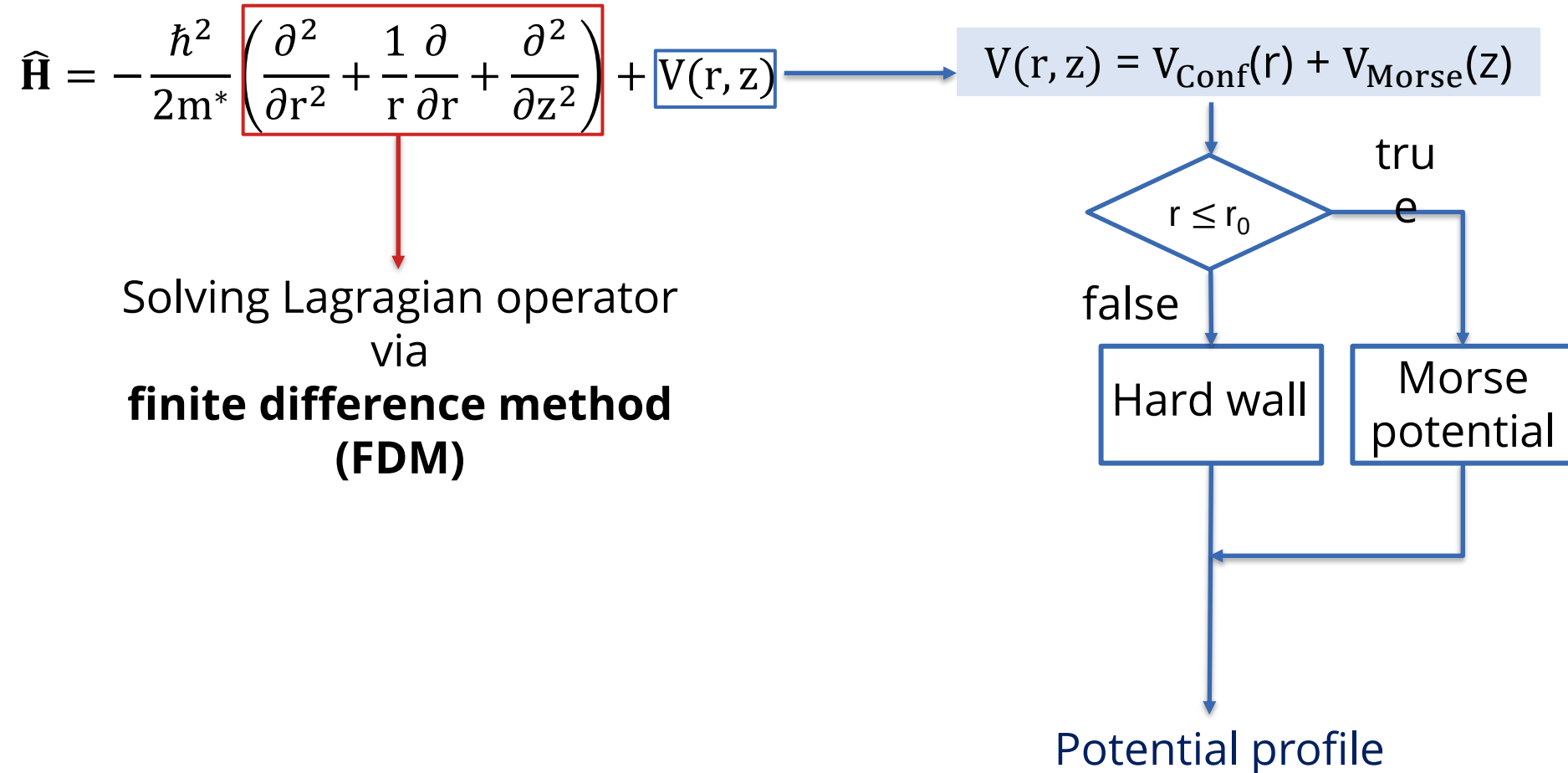


Degrees of freedom for the motion of He



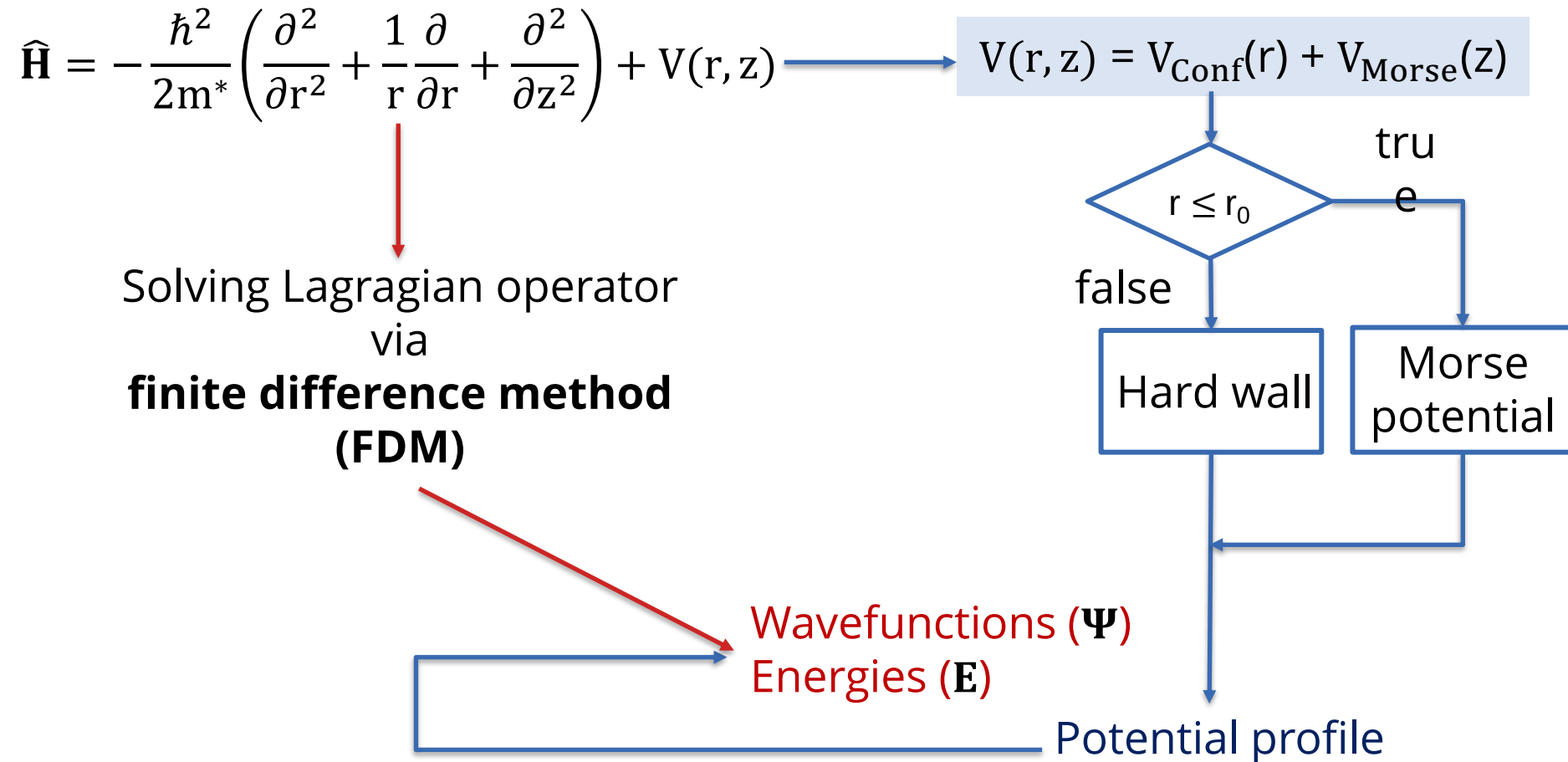
2D Schrödinger equations

$$\text{Schrödinger equations } \hat{H}\Psi = E\Psi$$



2D Schrödinger equation

$$\text{Schrödinger equation : } \hat{H}\Psi = E\Psi$$



Result: Morse potential

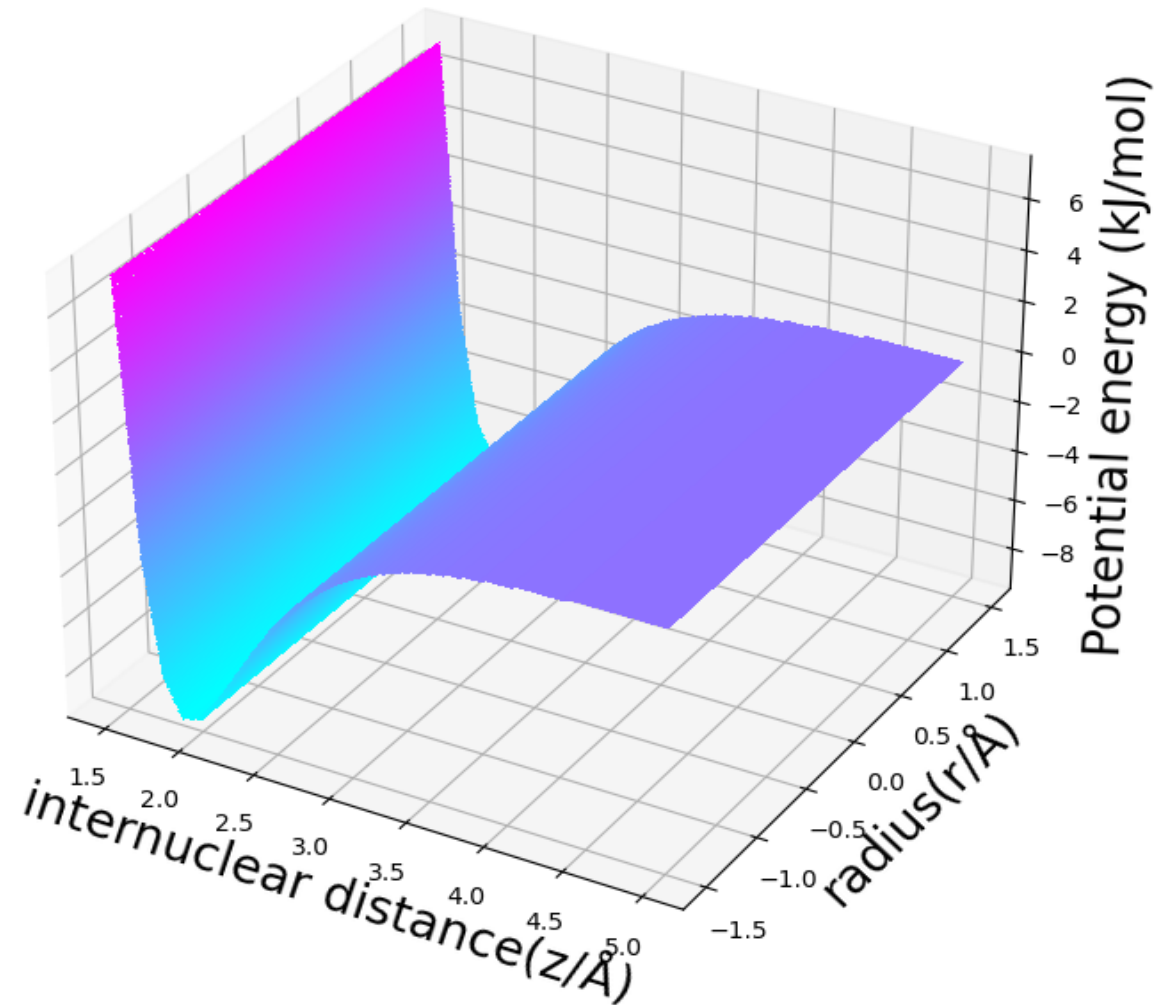
Range in Å:
[−1.5, 1.5] along the r-axis

Let's
set:

✓ $r_0 = r_{\max} = 1.5 \text{ Å}$

✓ $V_{\text{Conf}} = 0 \text{ kJ/mol}$

[1.5, 5] along the z-axis



Hard wall model

Range in Å :

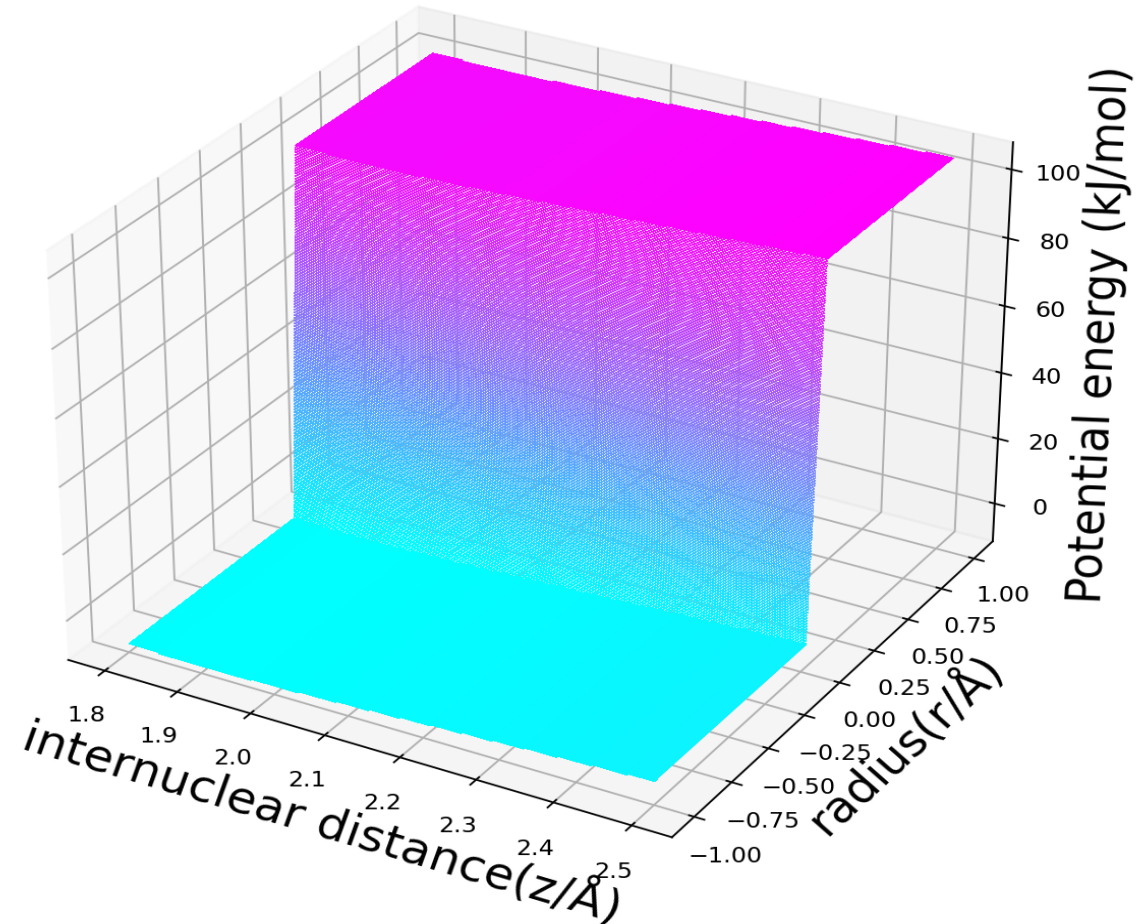
$[-1.5, 1.5]$ along the r-axis

Let's
set:

✓ $r_0 = 0.1 \text{ Å}$

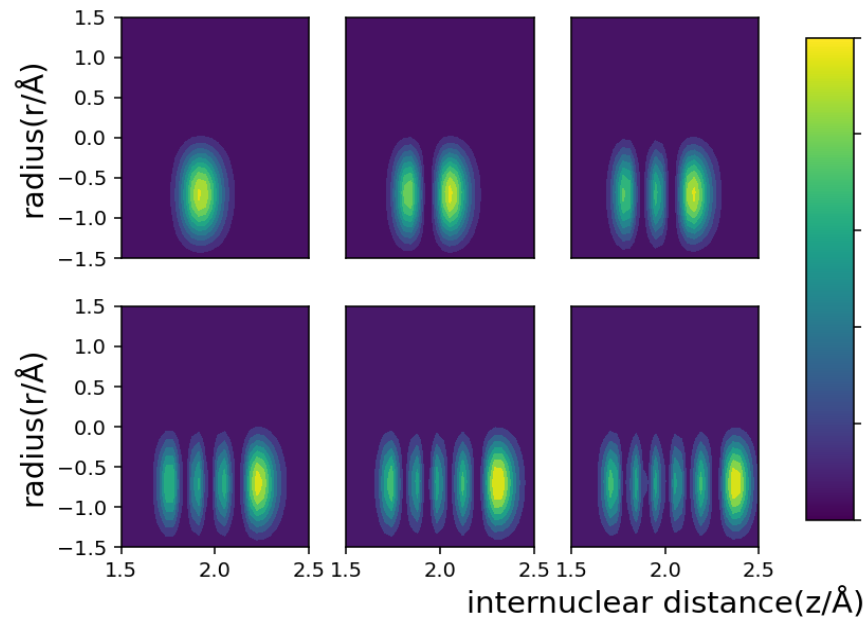
✓ $V_{Conf} \rightarrow 100 \text{ kJ/mol}$

$[1.8, 2.5]$ along the z-axis

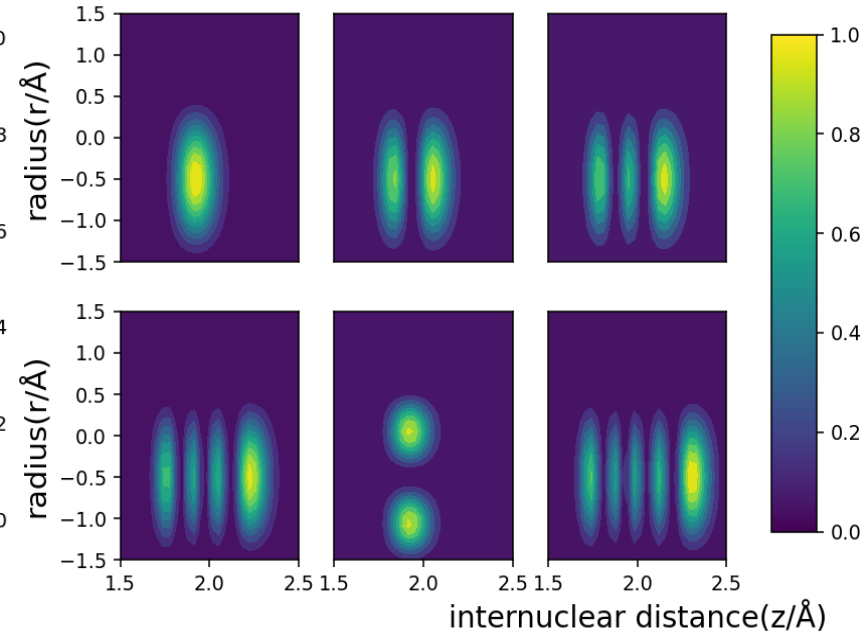


Wave functions

$r_0 = 0.1 \text{ \AA}$



$r_0 = 0.5 \text{ \AA}$



Range in $\text{\AA}$:

$[-1.5, 1.5]$ along the r -

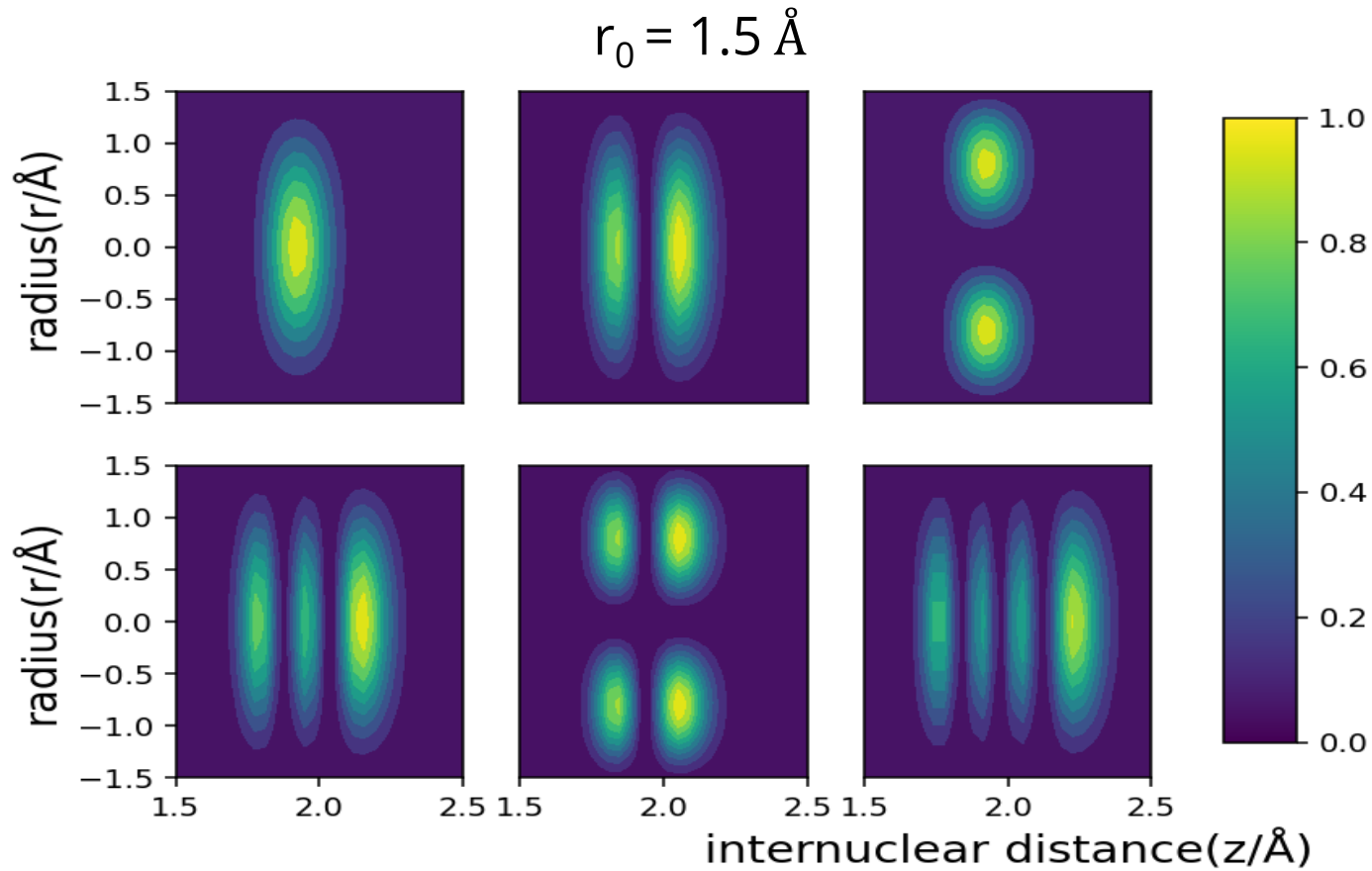
S

$[1.5, 2.5]$ along the z -axis

Total number of points :
 $N = 30 \times 30 = 900$ points

Figure 7: Plots of the square of the wave functions ($|\Psi|^2$) for a Morse potential with a confinement using Dirichlet boundary conditions for ($n_z=1,2,\dots$ and $n_r=1,2,\dots$). The color code bar applies to each of these plots

Wave functions



Range in $\text{\AA}$:
[$-1.5, 1.5$] along the r -axis
[$1.5, 2.5$] along the z -axis

Total number of points :
 $N = 30 \times 30 = 900$ points

Figure 8: Plots of the square of the wave functions ($|\psi|^2$) for a Morse potential with a confinement using Dirichlet boundary conditions for ($n_z=1,2,\dots$ and $n_r=1,2,\dots$). The color code bar applies to each of these plots

Energy levels

Table 2: Energy levels for different values of r_0

Numerical method (FDM)	Ground state E_0 (kJ/mol) for He	Ground state E_0 (kJ/mol) for ³He	Ground state E_0 (kJ/mol) for ⁴He	$\Delta ZPE_{4\text{He}/3\text{He}}$
$r_0 = 1.5 \text{ \AA}$	-5.114	-7.231	-7.549	0.318
$r_0 = 1 \text{ \AA}$	-5.109	-7.229	-7.547	0.318
$r_0 = 0.5 \text{ \AA}$	-5.084	-7.219	-7.539	0.32
$r_0 = 0.1 \text{ \AA}$	-5.027	-7.198	-7.523	0.325

ΔZPE Increase



Δr (step) = 0.08 ; Δz (step) = 0.12



Conclusion

Summary

- ❖ The system ($\text{Cu}^+\text{-He}$) allows to validate the QC and the numerical treatment of the interaction.
- ❖ ΔZPE between ^3He and ^4He gives us an idea about the potential ^4He / ^3He sieving.

Future plans

- ❖ Evaluate the influence of the confinement on the selectivity at a specific temperature $^{\circ}\text{C}$
- ❖ Numerical solution of the Schrödinger equation in 3D.
- ❖ Reproduction of adsorption data and selectivities from the literature.

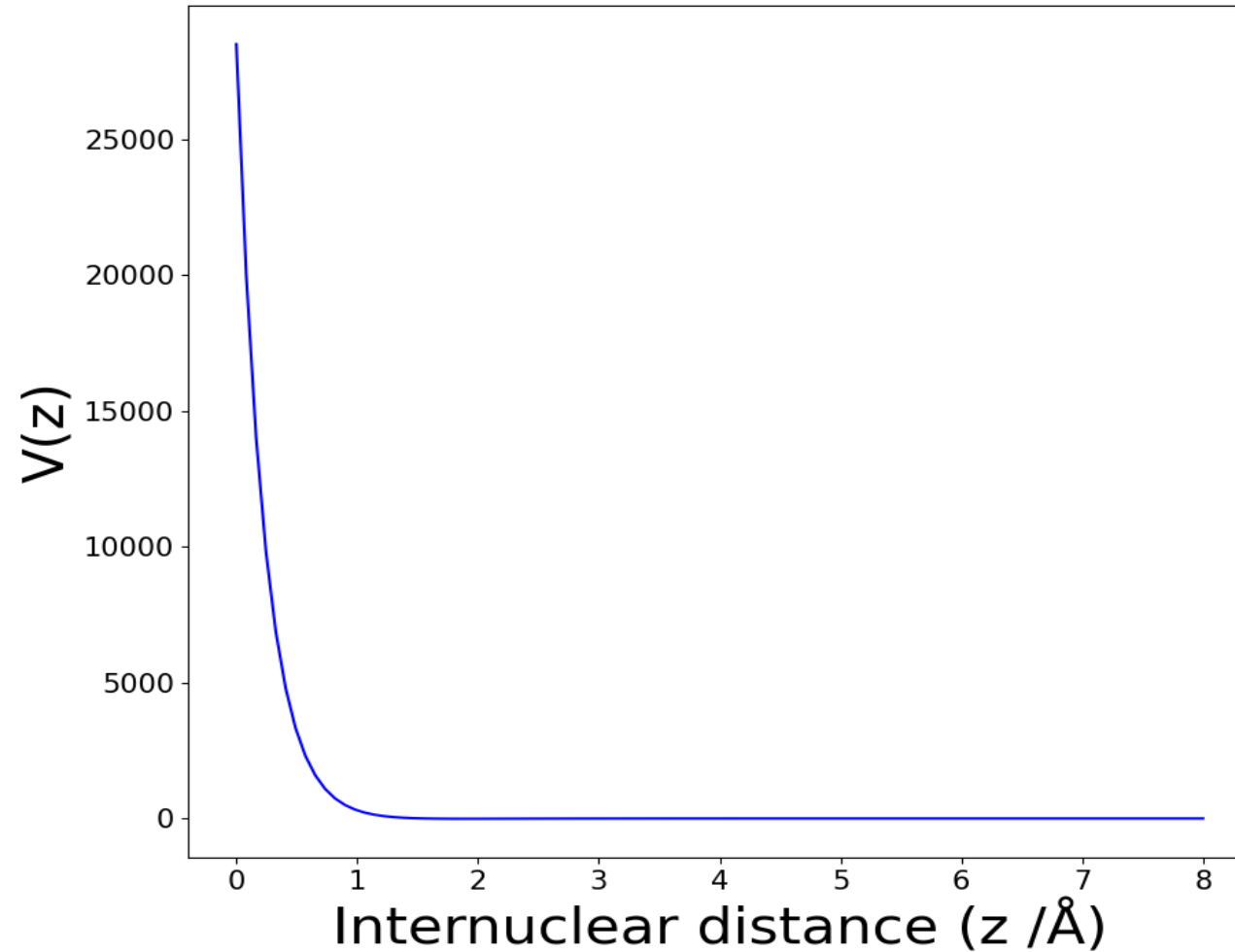
THANK YOU!



Backup slides



Morse potential when z start to 0



Anharmonicity: Energy levels

Morse potential : $V(z)=D_e \left[\left(1 - e^{(-a(z-z_e))} \right)^2 - 1 \right]$

Energy levels : $E_n = 2\pi\hbar\nu \left(n + \frac{1}{2} \right) - \frac{\left[2\pi\hbar\nu \left(n + \frac{1}{2} \right) \right]^2}{4D_e}$

where:
$$\begin{cases} \nu = \frac{a}{2\pi} \sqrt{2D_e/\mu} \\ n = 0, 1, 2 \dots \left[\lambda - \frac{1}{2} \right] \\ \lambda = \frac{\sqrt{2\mu D_e}}{a\hbar} \end{cases}$$

and ZPE : $E_0 = \pi\hbar\nu - \frac{1}{4D_e} \left(\frac{\pi\hbar\nu}{2} \right)^2$

n : vibrational quantum number

ν : state with a unit of frequency

D_e : dissociation energy

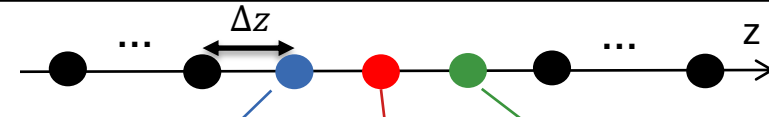
μ : reduced mass; $\mu = \frac{m_1 m_2}{m_1 + m_2}$
 ($m_1 = m_{cu}$; $m_2 = m_{4He}$ or m_{3He})

$\hbar$: reduced Planck constant $\left(\frac{h}{2\pi} \right)$

Finite Difference Method (FDM)

❖ 1D

$$\hat{H} = -\frac{\hbar^2}{2m^*} \frac{d^2}{dz^2} + V(z)$$



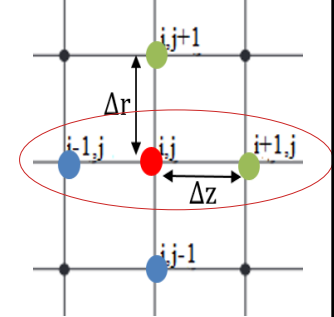
$$\frac{d^2\Psi(z_i)}{dz^2} = \frac{\Psi(z_i - h) - 2\Psi(z_i) + \Psi(z_i + h)}{(\Delta z)^2}$$

known ↑

❖ 2D

$$\hat{H} = -\frac{\hbar^2}{2m^*} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial z^2} \right) + V(r, z)$$

$$\frac{d^2\Psi(r_j)}{dr^2} = \frac{1}{r_j} \left(\frac{\psi_{i,j+1} - \psi_{i,j-1}}{2\Delta r} \right) + \frac{\psi_{i,j+1} - 2\psi_{i,j} + \psi_{i,j-1}}{(\Delta r)^2}$$



FDM...

Three dimensions

$$\underbrace{\left[-\frac{\hbar^2}{2m^*} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2} \right) + V(r, \theta, z) \right]}_H \psi(r, \theta, z) = E \psi(r, \theta, z) \quad (1)$$

Two axisymmetric dimensions :

❑ 2D- Axisymmetric structure ($\frac{\partial^2 \psi}{\partial \theta^2} = 0$)

❑ Symmetry boundary condition ($\frac{\partial \psi}{\partial r} = 0$)

❑ Singularity at $r = 0$

❑ The area is changing only in the radial direction : $dA = 2\pi dr$

Equation (1) can be writing as follows :

$$\left[-\frac{\hbar^2}{2m^*} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial z^2} \right) + V(r, z) \right] \psi(r, z) = E \psi(r, z) \quad (2)$$

$$\text{BCs : } \begin{cases} \text{left : } \psi(0, r) = 0 \\ \text{right : } \psi(L, r) = 0 \end{cases} \begin{cases} \text{bottom or top : } \psi(z, r_0) = 0 \\ \text{axial symmetry : } \left. \frac{\partial \psi(z, r)}{\partial r} \right|_{r=0} = 0 \end{cases}$$

Convergence

How to be sure that the cylindrical size is large enough?